

Adoption of Artificial Intelligence in Customer Relationship Management: Insights from Bangladesh's Digital Marketing Sector

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Abstract— The main objective of the study is to examine the effect of technological, organizational, and environmental factors, namely, PU of AI, OR, TC, and CP, and their interplay with each other on the adoption of AI-powered CRM in the context of digital marketing in Bangladesh. The quantitative research design was used with primary data being collected by using a structured questionnaire with a 5-point Likert scale. The marketing managers, digital marketing professionals, and CRM specialists were the targeted group and the analysis of 149 valid responses was conducted with the help of Smart PLS 4 and PLS-SEM to assess the hypothesis about the existing relationships. The results indicate that all the four determinants have a positive and significant impact on AICRM adoption, with CP having the strongest impact, and then OR, TC, and PU. The findings underscore the interactive effects between technology, organizational and environmental variables in determining the behavior to adopt AI, which justifies the relevance of the TOE framework in a developing country setting. In practice, the study is actionable in helping managers to improve AI adoption through investing in organizational preparation, harmonizing technology systems, and tracking competitor forces. The study highlights socially the prospects of AI adoption to enhance customer experiences and advance more effective digital marketing practices. It is innovative in that it incorporated several determinants in the framework of TOE in the specific case of AI-powered CRM adoption in Bangladesh. The constraints are the convenience sampling and cross-sectional data, which indicates that longitudinal studies should be used in further research.

Keywords— AI-powered CRM, digital marketing, AI adoption, TOE framework, organizational readiness, competitive pressure

I. INTRODUCTION

The advent of AI has brought digital marketing front to a new perspective, where a person can interact with the customers more personal and effectively. In the area of customer relationship management (CRM) the AI-driven systems will offer predictive analytics, automatic response, and information-based insight to enhance customer satisfaction and loyalty [1]. Businesses in Bangladesh are increasingly using AI based CRM technology as a competitive survival strategy, in marketing campaign streamlining and improving the interaction with customers. Welcoming AI to

CRM systems, the companies will be able to process vast amounts of client data in real-time and open the opportunity to communicate with the clients targeting and providing the service proactively [2], [3]. In addition, AI facilitates convenience of communication across channels, which is highly necessary in an interconnected society. The technological change is shifting the old CRM practices towards the more adaptive, intelligent and customer-focused practices as well [4]. Since then, customer relation management has been regarded as a key factor in business success, and it is related to acquisition, maintenance, and development of customer relations [5].

The conventional CRM was excessively reliant on manual processes and past data that limited its response and personalization capacities. With the advent of AI in digital marketing, machine learning, natural language processing, and predictive analytics have become components of CRM systems and have radically changed these dynamics [6], [7]. One of the areas that the Bangladesh Business are gradually discovering the importance of AI is in improving the customer experiences, streamlining operations and decision-making processes. In particular, AI-driven CRM will become significant within the sphere of this sort of business (retail, bank, and telecommunication) where the consumer requires the services to be custom-made [8]. The above development highlights the fact that the factors that drive the implementation of AI in the CRM practices must be known. Currently, the implementation of AI-based CRM in Bangladesh is growing but not proportionate, as larger firms adopt more advanced technology at a quicker rate compared to smaller ones (SMEs) [9]. Companies are using AI to customer segmentation, chatbots and sales prediction analytics to enhance the efficiency of their operations and establish customer interaction. However, this is accompanied by such challenges as the lack of technological infrastructure, lack of skilled specialists, and the problem of information privacy and unethical use of AI [10]. In addition to that, client awareness, and AI-based interaction plausibility remain at an early phase and influence the work of these systems. According to recent changes, AI uptake is increasing at a slow pace because of the competition, increasing digital literacy rates, and government-funded initiatives, which can be

viewed as a transition phase of the digital marketing landscape in Bangladesh [11].

Although digital marketing is increasingly becoming an important tool, most organizations in Bangladesh are finding it hard to exploit AI-based customer relationship management (AICRM) systems to improve customer engagement and operational efficiency. Although AI can be very useful, including through predictive analytics, personalized communication, and automated service, its implementation is frequently determined by such aspects as perceived usefulness (PU), the readiness of the organization (OR), compatibility (TC) with the technology, and competitive pressures (CP) [12], [13]. A common problem that many companies are struggling with is that they struggle to find balance between technological infrastructure, AI needs, skills of their staff, and how it reacts to the competition in the market. Besides, doubts about AI implementation into the current activities and perceived usefulness of the tools tend to have a negative effect on their implementation. The challenges are enhanced in the setting of Bangladesh, where digital transformation is a dynamic process, limited resources, the lack of technical expertise, and diverse degrees of management support. The overall impact of these determinants on the adoption of AICRM is vital to the organizations that wish to achieve competitive advantages and advance customer-oriented approaches.

RQ: How does PU, OR, TC, and CP affect the adoption of AI-powered CRM in Bangladesh in the digital marketing environment?

The objective of this study is to investigate the effect of the technological, organizational, and environmental factors on the adoption of AI-powered CRM within the Bangladesh digital marketing environment whereas focusing specifically on PU of AI, OR, TC, and CP factors. The present work is worthy because it helps to gain insight into what affects the implementation of AI-driven CRM in the digital marketing industry in Bangladesh. The research can assist the manager and decision-makers by identifying the role of PU, OR, TC, and CP, which will help them to implement AI solutions more appropriately. The study also adds to the academic informational base, as it provides the application of the TOE framework in a developing country setting to guide the further research and other practical implementations in the process of customer engagement and competitive advantage improvement.

II. THEORETICAL FOUNDATIONS AND HYPOTHESES DEVELOPMENT

The research is based on the TOE framework that presents an extensive prism to review the variables that affect the process of adopting innovation in the organizational context [14]. TOE framework is specifically appropriate to the adoption of AI-powered CRM since it takes into account three important contexts, that is, technological, organizational, and environmental. The technological environment outlines the influence of PU and TC on the decision to implement AI systems. The organizational setting demonstrates the contribution of OR such as resources, skills and management support towards adoption. Lastly, the environmental context is the external force, like CP, that compels organizations to innovative solutions. TOE can be used to investigate the multi-dimensional determinants of AICRM adoption in Bangladesh in a systematic way to gain theoretical and

practical understanding of how internal capabilities and external pressures interact [15].

PU represents how much people think that the implementation of AI-powered CRM would allow them to work better and have better interactions with customers. The research has demonstrated that employees and managers can be more likely to use AI tools when they believe that these tools can help in automating the processes, personalizing communication, and gaining actionable insights [16], [17]. The PU is an important factor of AI adoption in the context of digital marketing in Bangladesh because AI can be used to optimize the engagement with customers and facilitate the decision-making process. This is in line with the technological aspect of the TOE framework that focuses on how the attributes of innovation affect the decisions to adopt innovations. This hypothesis is, therefore, formulated in order to investigate the positive impact of PU on AICRM adoption.

H1: PU of AI positively influences the use of AI-based CRM in digital marketing.

OR can be defined as the availability of the resources, infrastructure, and competent people that a firm has the capabilities to carry out the implementation of AI-powered CRM. Better-equipped companies are more prepared to introduce new technologies into the current systems, educate the staff, and facilitate constant enhancement of the system [17], [18]. OR makes sure that the digital marketing industry in Bangladesh is not hesitant about its implementation due to the lack of resources or managerial support, which is essential to the successful implementation of AI. This is conciliated with the organizational aspect of the TOE framework that underscores the fact that internal readiness and capability are the driving force in the adoption decision. Consequently, this is the hypothesis that is formulated to investigate the beneficial role of OR in AICRM adoption.

H2: OR positively influences the use of AI-based CRM in digital marketing.

TC is the level to which AI based CRM systems match the current technological infrastructure and processes involved within an organization. Effectively integrating new AI tools with existing systems will enable companies to minimize the difficulties of implementation, reduce disruptions, and enhance their efficiency [19], [20]. Within the scope of digital marketing in Bangladesh, compatibility is important in ensuring a seamless adoption process because companies are usually constrained in terms of IT facilities and outdated systems. This aspect is in line with the technological aspect of the TOE framework which lays stress on the fact that the uptake of innovations depends on their compatibility with the available resources and work processes. Therefore, this hypothesis is formulated to determine whether TC affects the adoption of AICRM positively.

H3: TC positively influences the use of AI-based CRM in digital marketing.

CP would indicate how firms are under pressure by their competitors to implement innovative technologies to sustain or improve their position in the market. With the competition using AI-powered CRM to enhance customer engagement and operating performance, organizations tend to adopt the technology [21], [22]. The growing competition in the digital marketing environment of Bangladesh encourages companies to embrace AI-based solutions in order to distinguish

themselves, maintain their customers and get a competitive edge. This goes in line with the environment aspect of the TOE framework that brings out the role of the external forces of the market in the decision of technology uptake. Thus, the hypothesis is created to examine the beneficial role of CP toward the adoption of AICRM.

H4: CP positively influences the use of AI-based CRM in digital marketing.

Based on the above discussion and the hypotheses, a conceptual framework has been developed in Fig 1:

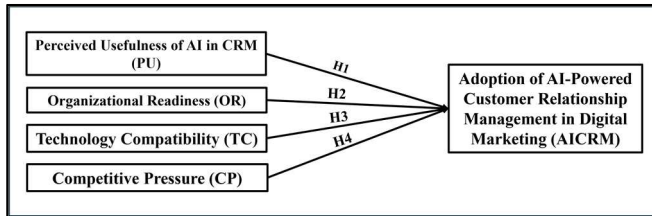


Fig. 1. Conceptual framework

III. METHOD

The study adopted a quantitative research approach in order to investigate the predictors of AI-based CRM adoption in the digital marketing environment in Bangladesh. The use of quantitative research was suitable since it would enable us to measure and statistically analyze the association of variables. The structured questionnaire was used to collect primary data, the questionnaire was developed on a 5-point Likert scale (strongly disagree to strongly agree). The marketing managers, digital marketing practitioners, and CRM specialists at firms in Bangladesh were included in the target population because they have direct experience in CRM decision-making. The individual respondent, who was the representative of his or her respective organizations, was taken as the unit of analysis. The method of convenience sampling was applied because of the availability and time limitation and 200 participants were given questionnaires. One hundred and eighty responses were received and after filtering out those that were either unfinished or those that were not consistent, 149 responses were retained to be analyzed and this generated a satisfactory error margin. To make the questionnaire clear, relevant, and appropriate in terms of items, the questionnaire was pre-tested on a group of five professionals working in digital marketing and AI to make sure that the questions were clear, relevant, and appropriate. Expert review and conformity to the previous literature ensured the content validity and construct validity. Face-to-face interaction and online survey were used to gather the data. The answers were coded and inputted into Smart PLS 4 software to be analyzed. The hypothesized relationships were tested using PLS-SEM because it is suitable in a small to medium sample size and predictive modeling [23]. The use of this statistical method was justified because it enables the simultaneous measurement and structural model to be assessed, and the multifaceted relationship between latent constructs is easily studied. In Table I, the reliability results indicate that each construct achieved a Cronbach's Alpha value greater than 0.80, which is generally considered acceptable and even strong for social science research [24].

TABLE I. RELIABILITY OF THE MEASUREMENTS

Construct	Items	Cronbach's Alpha
PU	4	0.83

OR	4	0.85
TC	4	0.81
CP	3	0.87
AICRM	3	0.81

IV. RESULTS

The demographic profile of the 149 respondents that were used in this study is presented in Table II. Most of the participants were male (61.1%), and female respondents were 38.3% which means that more males were represented in the population under survey. The majority of participants were between 20 and 29-year-old (40.9%) and 30-39-year-old (34.2%) indicating that the digital marketing and AI sectors have relatively young labor force. In terms of employment status, the highest percentage of 42.3 was in mid-level employees, followed by entry level (32.9%) and senior level employees (24.8%) as they represent different levels of organizational experience. The advertisement firms (36.9%) and IT (20.8%) industries were the most common in the sample, which indicates the applicability of AI in these sectors. The monthly household income was also clustered to the 20,000-50,000 range (45.0%), which implies that respondents are also characterized by a moderate level of income. Altogether, Table II shows a sample representative of professionals that will be pertinent to the deployment of AI-powered CRM in Bangladesh

TABLE II. DEMOGRAPHIC PROFILE OF THE RESPONDENTS

Variable	Category	Frequency	Percent (%)
Gender	Male	91	61.1
	Female	57	38.3
Age	20-29 Years	61	40.9
	30-39 Years	51	34.2
	40-49 Years	37	24.8
	50-59 Years	10	6.7
Current Employment Status	Entry-level	49	32.9
	Mid-level	63	42.3
	Senior-level	37	24.8
Industry Sector	Information Technology	31	20.8
	Advertising Firm	55	36.9
	Retail	21	14.1
	Banking/Finance	19	12.8
	Others	23	15.4
Monthly Household Income	Below 20,000 BDT	43	28.9
	20,000-50,000 BDT	67	45.0
	50,001-100,000 BDT	27	18.1
	Above 100,000 BDT	12	8.0
Total		149	100

The findings of the measurement model to evaluate reliability and validity of the constructs are observed at Table III. The item loading was above the suggested level of 0.70, which is an indication of high reliability of the indicator [23]. AVE values of the constructs were between 0.64 and 0.79 and this is more than the necessary convergent value of 0.50 and this establishes a convergent validity. CRs were between 0.87 to 0.92 which is higher than the suggest 0.70 CR, and show high internal consistency of the constructs. The reliability and validity of PU, OR, TC, CP, and Adoption of AI-Powered CRM were high and guaranteed the fact that constructs reflect the corresponding theoretical ideas. On the whole, Table III proves the strength of the measurement model and its ability to be analyzed further structurally.

TABLE III. MEASUREMENT MODEL

Constructs	Items	Loading	AVE	CR
PU of AI in CRM	PU1	0.82	0.66	0.89
	PU2	0.81		
	PU3	0.78		
	PU4	0.84		
OR	OR1	0.86	0.70	0.90
	OR2	0.87		
	OR3	0.81		
	OR4	0.80		
TC	TC1	0.79	0.64	0.87
	TC2	0.81		
	TC3	0.80		
	TC4	0.79		
CP	CP1	0.91	0.79	0.92
	CP2	0.90		
	CP3	0.85		
AICRM	AICRM1	0.87	0.73	0.89
	AICRM2	0.87		
	AICRM3	0.82		

Table IV shows the Fornell-larcker criterion to determine the discriminant validity between the constructs. The results indicated based on the square root of the AVE of the constructs (the diagonal values) that the correlations of the construct with the rest of the constructs was smaller than its square root of the AVE, which demonstrates that there is sufficient discriminant validity [25]. As an example, the square root of AVE with PU (0.81) is greater in comparison to its relationships with OR (0.67), TC (0.67), CP (0.71) as well as AICRM (0.73). Likewise, all other constructs: OR, TC, CP and AICRM had higher diagonal values as compared to off-diagonal correlations which proved that each construct is unique to the other. These findings support the idea that the measurement items have a high measure of the intended constructs without any major overlap and hence the strength of the model in the structural analysis. Table IV, in general, substantiates the fact that the discriminant validity was attained in the current research.

TABLE IV. FORNELL-LARKER CRITERION

Construct	PU	OR	TC	CP	AICRM
PU	0.81				
OR	0.67	0.83			
TC	0.67	0.80	0.80		
CP	0.71	0.83	0.80	0.89	
AICRM	0.73	0.82	0.80	0.87	0.85

Table V shows the results of testing hypothesis through PLS-SEM. The four hypothesized relationships were all supported, which confirms the existence of strong positive influences of the independent variables on the adoption of AI-powered CRM. CP was the most influential variable on AICRM ($T=4.96$, $p<0.01$), which implies that the outside market factors are considered crucial in implementing AI. The effect of OR was also significant ($T=2.55$, $p<0.05$), indicating that internal resources and preparedness are as well important. PU and TC had positive relationships with AICRM adoption ($T=2.27$ and 2.36 , $p<0.05$ respectively), which validates the perceived benefits and compatibility to the current technology are the critical factors. These findings speak in favor of the TOE framework, proving that all of the three components, i.e., technological, organizational, and environmental ones, have a combined effect on the adoption of AI in digital marketing.

Table V confirms all the hypotheses proposed in it making them both theoretical and practical.

TABLE V. HYPOTHESIS TESTING RESULTS

Hypothesis	Relationship	T Statistics	P Values	Decision
H1	CP → AICRM	4.96	0.00	Accepted
H2	OR → AICRM	2.55	0.01	Accepted
H3	PU → AICRM	2.27	0.02	Accepted
H4	TC → AICRM	2.36	0.02	Accepted

Table VI shows the coefficient of determination (R^2) of the dependent variable, Adoption of AI-Powered CRM. The R^2 value of 0.81 shows that the independent variables PU, OR, TC, and CP explain 81 percent of the variance in AICRM adoption. The adjusted R^2 of 0.80 is another indicator that indicates the strength of the model since it has considered the number of predictors implying that it would do very little overestimation. This is a strong predictive power, which proves that the determinants chosen are powerful predictors of AI adoption in the digital marketing environment of Bangladesh. In general, Table VI confirms that the proposed model is effective in the process of capturing the major factors that have a significant effect on AICRM adoption, which is in line with the TOE theoretical framework.

TABLE VI. COEFFICIENT OF DETERMINATION (R^2)

Construct	R^2	Adjusted R^2
AICRM	0.81	0.80

Fig. 2 shows that the independent constructs of Adoption of AI-Powered CRM have a value of $R^2=0.81$, which implies that the independent constructs of PU, OR, TC, and CP jointly contribute to the adoption of AICRM at a level of 81 percent. This is regarded as a very high explanatory power, which implies that the model has a great predictive power in adoption behavior. CP (0.45) has the largest effect among the predictors, then OR (0.22), TC (0.17), and PU (0.15) have the relatively smaller nonetheless positive effects. The high R^2 shows that these four factors have a significant influence on decisions of adoption or not in organizations and the competitive forces have the most important influence. In this way, the model is statistically sound and has a practical meaning, as it emphasizes both external and internal preparedness to propel AI-CRM integration.

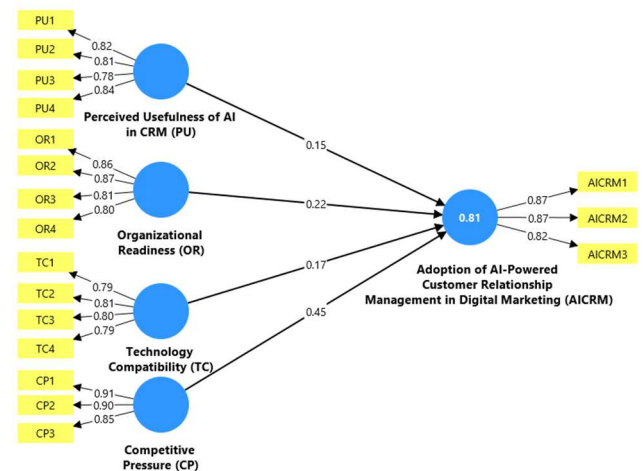


Fig. 2. Structural Equation Modelling and Path Coefficient.

V. DISCUSSION

The current paper has examined the factors that determine the implementation of AI-based CRM in the digital marketing environment of Bangladesh. This study employed the theoretical approach of TOE to explore the impact of PU, OR, TC, and CP on the embrace of AI. It has been discovered that all the four factors have a significant positive effect on the adoption of AICRM, in that order, CP is the strongest, then OR, TC, and PU. These findings underline the interactive impact of technological, organizational, and environmental settings on the adoption decisions of firms and that internal capabilities as well as external market force have a significant role to play to make AI adoption successful [14], [18]. The model has a high explanatory power ($R^2 = 0.81$) which validates the fact that the determinants used together have been able to explain a large share of variance in AICRM adoption which is a strong indicator that they have a good understanding of the adoption behavior in this context.

A. Contributions of the study

Theoretically, this research study has provided its contribution by expanding the concept of TOE framework to the new field of AI-driven CRM in a developing country context. Although the adoption of AI in the technologically advanced settings has been the focus of previous studies, this paper shows that TOE is applicable to the issue of adoption in Bangladesh, meaning that the environmental pressures and the technological and organizational readiness factors play an important role [17], [26]. The research offers evidence-based information on how companies can use AI to improve customer relationship, efficiency of marketing activities, and competitive advantage to fill the gap in localized research in the Bangladeshi digital marketing environment. Practically, the findings offer a practical recommendation to the managers and policymakers. Cultivating internal capabilities, TC, and organizational culture open to AI should be the core focus of the firms, and the competition on the market should be tracked to implement AI-based CRM systems strategically. This paper indicates that it is also important to learn about PU and determine the fit between technology and current workflows to ensure that the benefits of adoption are maximized. Regarding novelty, the study is unique in that it incorporates various determinants in the TOE framework to examine the adoption of AI-powered CRM in particular to the context of Bangladesh, which can serve as a model that can be transferred to other emerging economies. On the whole, the research is both theoretically, empirically, and practically valuable, providing a clear picture of the AI adoption drivers, justifying the strategic decision-making process, and creating the path to future research of AI-enabled digital marketing solutions [16], [22].

VI. CONCLUSION

The study has dealt with the acute problem of poor uptake of AI-based CRM systems by the companies in the digital marketing industry in Bangladesh. Although AI could have certain advantages in promoting customer interactions, facilitating marketing operations, and increasing operational efficiency, most organizations have difficulties with TC, OR, PU, and CP. Exploring these determinants in terms of TOE framework, the study offered a systematic insight into the aspects that influence the adoption of AICRM in a developing country setting. The results indicate that CP has the greatest impact on AICRM adoption, and the second, third, and fourth

in influence on adoption are OR, TC, and PU that describe 81% of the variance of adoption behavior. These findings underscore the importance of the interaction between the internal features and the external environmental forces in spurring the adoption of AI. Theoretically, the research applies the TOE model to the AI-driven CRM situation in the emerging markets, which proves its relevance to the more technologically sophisticated environment. In practice, the study provides managers with practical implications, providing recommendations to invest in organizational resources, training employees, integrating technology, and being competitive to gain the most out of AI application. To carry out future research, longitudinal studies would be appropriate to investigate the changes in the adoption of AICRM through the years and determine the effects on customer satisfaction and firm performance in the long-term. The researchers can also investigate other determinants like the issue of data privacy, moral issues and user trust, or use the model to test the adoption patterns in other developing markets. Also, incorporation of qualitative methods may give more insight into the organizational and cultural aspects that affect AI adoption. In general, this paper provides a starting point to scholars and practitioners to identify and support the successful implementation of AI-powered CRM to foster innovation and competitive edge in online marketing.

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